Introduction & Hook

“When you enter a VR game or simulation, what creates the world around you?”

Explain:

VR environments are not random images — they are carefully modeled digital worlds made of:

- Geometry
- Textures
- Lighting
- Objects
- Physics

Today we study how those worlds are constructed.

🌍 PART 1: Environment Modeling

What is Environment Modeling?

Definition:

Environment modeling is the process of creating the virtual surroundings in a VR system.

It includes:

- Terrain
- Buildings
- Sky
- Vegetation
- Roads
- Interior spaces

Types of Environment Modeling

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1. Geometry-Based Modeling

Using:

- Vertices
- Edges
- Polygons

Example:

- Creating a room using cubes and planes
- Building a city with 3D meshes

Used in:

- Unity
- Unreal
- Blender

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2. Procedural Modeling

Automatically generating environments using algorithms.

Example:

- Infinite terrain generation
- Procedural forests
- Random city generation

Advantage:

- Large worlds with less manual effort

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3. Photogrammetry-Based Modeling

Creating environment using real-world photos.

Example:

- Scanned monuments
- Real-world landscapes

High realism but heavy performance cost.

Environment Components in VR

A VR environment contains:

- Static objects (walls, terrain)
- Dynamic objects (characters)
- Light sources
- Sound sources
- Physics zones

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PART 2: Object Representation in VR

How Objects are Represented

In VR, objects are represented using:

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1. Mesh Representation

A mesh consists of:

- Vertices (points)
- Edges (lines)
- Faces (triangles)

Triangles are most important in rendering.

Example:
A cube = 12 triangles.

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2. Material and Texture

Object representation includes:

- Color
- Texture maps
- Normal maps
- Reflectivity
- Transparency

Example:
A wooden table uses:

- Wood texture
- Surface shading
- Reflection settings

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3. Transform Properties

Each object has:

- Position
- Rotation
- Scale

Stored inside scene database.

Object Hierarchy and Attributes

Objects in VR also contain:

- Collision properties
- Physics properties (mass, gravity)
- Animation data
- Interaction scripts

Example:

A VR door object includes:

- Mesh
- Material
- Hinge physics
- Open/close script

Object representation is more than shape — it includes behavior.

Environment vs Object (Clarification)

Environment	Object
Overall world	Individual entity
Contains terrain, sky	Chair, table, avatar
Mostly static	Can be static or dynamic
Large scale	Small scale

Environment = Stage
Objects = Actors

Recap + Questions

Recap:

- Environment modeling creates the virtual world
- Objects are represented using meshes and materials
- VR objects include geometry + physics + behavior
- Modeling choice affects realism and performance

Quick Oral Questions:

- What is environment modeling?
- What is a mesh?
- Why are triangles important?
- What is difference between environment and object?